TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

Reminders
System of inequalities: graph and shade each one; the overlap is the solution set.
A solution makes EVERY inequality true. Points on a dashed line are not solutions.
In a story: no negative amounts, so stay in the first quadrant.

1. Which system has no solution?

Tip: Same slope? Check which sides are shaded.

A. $y > 2x + 3$ and $y > 2x - 1$

B. $y < 2x + 3$ and $y > 2x - 1$

C. $y > 2x + 3$ and $y < 2x - 1$

2. A workout is at most 40 minutes and burns at least 300 calories. Walking (w) burns 5 calories a minute and running (r) burns 10. Which (w, r) is a solution?

Tip: Check the minutes AND the calories.

F. (10, 30)

G. (30, 5)

H. (20, 25)

3. Write the system of inequalities for the workout in Question 2.

> **Tip:** One inequality for minutes, one for calories.

Minutes: $w + r$ ____ 40

Calories: $5w + 10r$ ____ 300